5.03 Continuous Random Variables			
5.03a	Continuous random variables		a) Understand and be able to use the concept of continuous random variable, a probability den function (p.d.f.) and a cumulative distribution f (c.d.f). *Includes the normal, continuous uniform and expone distributions.* *Includes understanding informally the link between t exponential and Poisson distributions.* *Includes knowing and being able to use the formula f mean and variance of the continuous uniform and ex distributions.* *For the underlying content on normal distributions, s sections 2.04e, 2.04f and 2.04g.*
5.03b	Probability density functions		b) Be able to use a probability density function (i where defined piecewise) to solve problems in probabilities. *Includes knowing and being able to use* $\int_{-\infty}^{\infty} \mathrm{f}(x)\,\mathrm{d}x =$
5.03c			c) Be able to calculate the mean and/or variance distribution using the formulae $\mu = \mathrm{E}(X) = \int$ and $\sigma^2 = \mathrm{Var}(X) = \int (x-\mu)^2 \mathrm{f}(x)\,\mathrm{d}x = \int x^2 \mathrm{f}(x)\,\mathrm{d}x$
5.03d			d) Be able to use the general result $\mathrm{E}(\mathrm{g}(X)) = \int \mathrm{g}(x)\mathrm{f}(x)\,\mathrm{d}x$, where $\mathrm{f}(x)$ is the prob density function of the continuous random var and $\mathrm{g}(x)$ is a function of X.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
5.03e	Cumulative distribution functions		e) Be able to find and use a cumulative dist function (including where defined piecev problems involving probabilities. *Includes being able to use* $\mathrm{F}(x) = \mathrm{P}(X \le x) = \int$
5.03f			f) Know and be able to use the relationship probability density function, $\mathrm{f}(x)$, and the distribution function, $\mathrm{F}(x)$, and use eithe the median, quartiles and other percenti
5.03g			g) Be able to find and use the cumulative di functions of related variables. *e.g. Given the c.d.f. of* X*, find the c.d.f. of* Y *and p.d.f. of* Y *where* $Y = X^3$.